Curriculum vitae : Arnaud HUBERT, PhD

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Education

Phd in Computational neuroscience, Modelling endocannabinoid-mediated synaptic plasticity and its implication in fast learning.
Nov 2022 - Mar 2026, AistroSight team, INRIA, France

Working on neuropixel data & SpikeSorted data, SpikeSorting, Modeling, Programming, Mean field model

Neuromatch Academy: computational neuroscience
Jul 2024 (3 weeks), remote online classes

Model Fitting, Generalized Linear Models, Dimensionality Reduction, Deep Learning, Linear Systems, Biological Neuron Models, Dynamic Networks, Bayesian Decisions, Hidden Dynamics, Optimal Control, Reinforcement Learning, Network Causality & Graduation.

Master graduation internship at INRAE
Apr 2022 - Sep 2022, INRAE, France

Programming with R, Working on satelitte spatial data, Gaussian free field.

Master's degree: Applied matematics for biology and medecine
2020 – 2022, Claude Bernard University – Lyon 1, France

Statistics, Stochastic process, Analysis.

Scientific production & communication

Papers

Nature Neuroscience (in publish), « Striatal endocannabinoids drive one-shot learning »,
Co-author with our colaborators from Laurent Venance’s team.
https://hal.science/hal-05574724v1/file/Piette-et-al-Nature-Neurosci-revised2-17032026.pdf

Paper Communication,WASET « Reconstructing Large-Scale Heterogeneous Quadratic-Integrate-and-Fire Networks Using Their Dominant Modes »
Main author, 2026
https://publications.waset.org/10014560/reconstructing-large-scale-heterogeneous-quadratic-integrate-and-fire-networks-using-their-dominant-modes

Repository

Synacomp, a Python package for applying arbitrary models to spike sorted data.
https://gitlab.inria.fr/aistrosight/synacomp

Spikeinterface-regions, A Python package to assign brain regions to SpikeInterface object
https://gitlab.inria.fr/ahubert/spikeinterface-regions

Communication

Poster, "Exploring behavioral correlations with neuron activity through synaptic plasticity."
Presented at the Bernstein conference 2024.
https://gitlab.inria.fr/ahubert/poster-bernstein-conference-2024

Teaching

Seminar teacher

2023 – 2025, Claude Bernard University – Lyon 1, France
50 hours of Algebra – D1.
6 h of Calculus – D1.
6 hours of probability – D3.

Student tutoring,

Volunteer help to post graduate student in maths.
Private teaching in maths and physics. Mostly final year level.

Skills

Languages

French : native
English : B2

Programming

Python (see https://gitlab.inria.fr/ahubert)
R (Bachelor's degree)